

National Construction Cost Database (NCCDB)

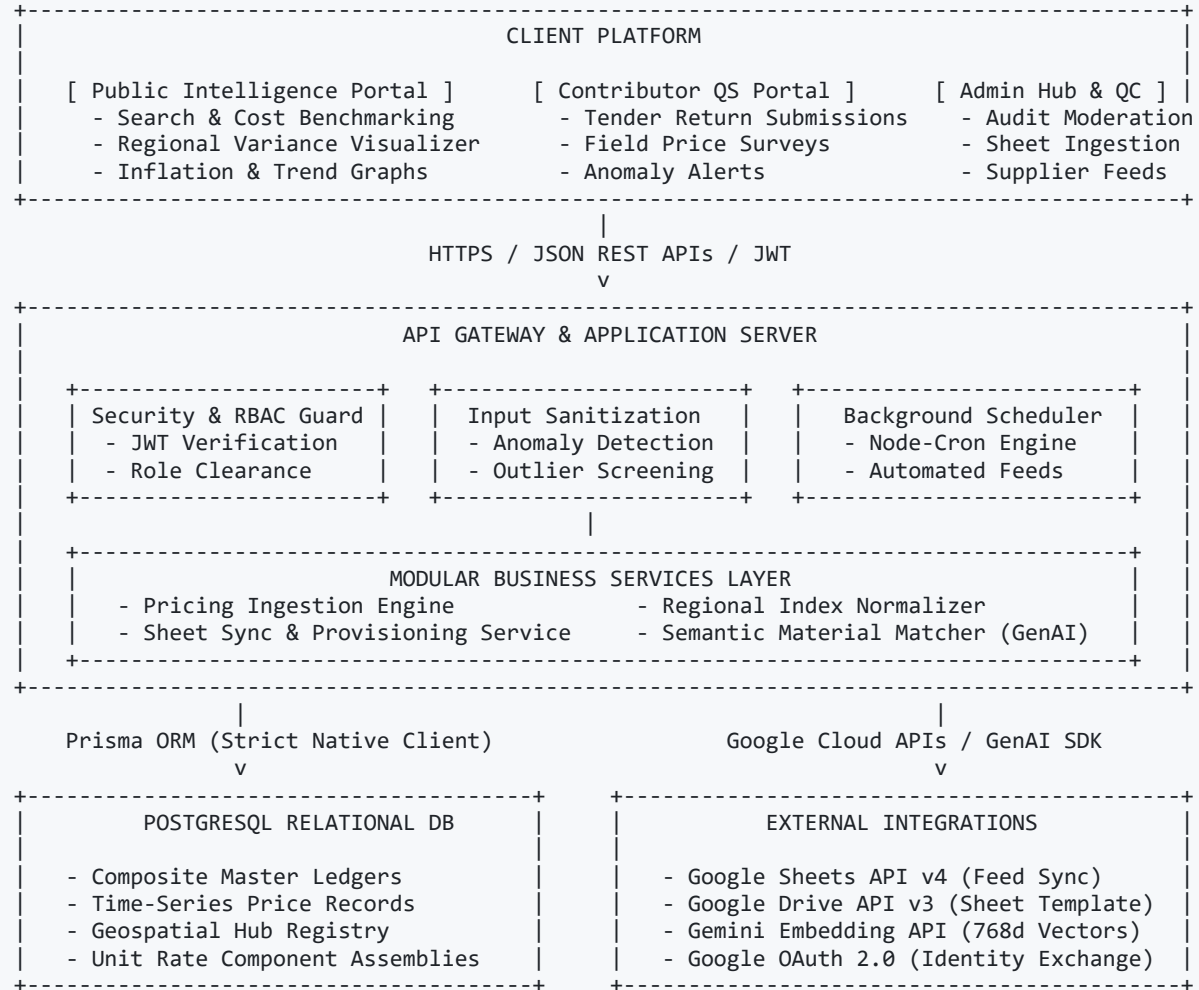
Technical Architecture & Data Platform Specification

Executive Summary

The **National Construction Cost Database (NCCDB)** is an enterprise-grade digital market intelligence and cost benchmarking platform engineered for the construction and civil engineering industry. It ingests, standardizes, validates, and models construction material prices, labor wage rates, equipment plant costs, and composite unit rate assemblies across geopolitical zones and primary industrial hubs.

This document details the architectural foundation, relational schema, data ingestion lifecycles, and analytical pipelines powering NCCDB. It is designed to walk technical stakeholders, founders, and engineering leaders through the core platform design without exposing proprietary secrets or operational credentials.

NCCDB HIGH-LEVEL PLATFORM TOPOLOGY



1. System & Tech Stack Overview

Dimension	Architectural Choice	Technical Rationale & Role
Core Runtime	Node.js (ES Modules) via TypeScript (~5.8.2) & tsx engine	Strict static typing, native ESM module resolution, and asynchronous non-blocking event-driven processing.
Backend Framework	Express.js (4.21.x)	Lightweight, decoupled HTTP routing, minimal overhead for REST endpoints, extensible middleware pipelines.
Data Access & ORM	Prisma Client (6.19.x)	Type-safe query building, declarative migrations, connection pooling, compile-time relational validation.
		ACID-compliant relational storage fixed-

2. Database Architecture & Entity Relationships

The data layer is modeled around three interdependent operational pillars:

1. **Core Market Intelligence Ledger:** Tracks physical materials, categories, location hubs, and historical price records with strict moderation workflows.
2. **Unit Rate Assembly System:** Powers parametric Bill of Quantities (BOQ) estimations by breaking down work items into material constants, plant equipment, and labor rates with configurable markups.
3. **Master Semantic Registry:** Normalizes disparate supplier nomenclature into standard catalog codes using vector embeddings and alias lookup tables.

2.1 Entity Relationship Diagram (ERD)

```
erDiagram
    %% Core Identity & Profiles
    users ||--o{ supplier_profiles : "has profile"
    users ||--o{ price_records : "submits"
    users ||--o{ price_records : "approves"

    %% Material & Taxonomy
    material_categories ||--o{ materials : "categories"
    materials ||--o{ material_suppliers : "supplied by"
    supplier_profiles ||--o{ material_suppliers : "supplier"

    %% Price Record Engine
    materials ||--o{ price_records : "priced as"
    locations ||--o{ price_records : "located at"

    %% Unit Rate Composition Engine
    unit_rate_items ||--o{ unit_rate_components : "contains"
    materials ||--o{ unit_rate_components : "serves as material component"
    labor_plant_library ||--o{ unit_rate_components : "serves as labor or plant component"

    %% Master Semantic Registry
    master_materials ||--o{ material_aliases : "maps to aliases"

    users {
        string id PK
        string email UK
    }
```

2.2 Relational Entity Catalog

Table: `price_records` (Core Market Transactions)

The primary ledger capturing point-in-time cost data points across Nigeria's construction markets.

Column	Data Type	Modifiers / Constraints	Description
<code>id</code>	<code>TEXT</code> (UUID)	<code>PRIMARY KEY</code> , Default: <code>gen_random_uuid()</code>	Unique immutable log record identifier
<code>materialId</code>	<code>TEXT</code>	<code>FOREIGN KEY</code> → <code>materials(id)</code> <code>ON DELETE CASCADE</code>	Reference to catalog material
<code>locationId</code>	<code>TEXT</code>	<code>FOREIGN KEY</code> → <code>locations(id)</code> <code>ON</code>	Reference to geographical market

Table: **materials** (Catalog Taxonomy)

Canonical catalog entries representing physical materials specified in Bills of Quantities.

Column	Data Type	Modifiers / Constraints	Description
id	TEXT (UUID)	PRIMARY KEY	Unique material identifier
name	TEXT	NOT NULL	Standard trade description
specification	TEXT	Nullable	Technical standard compliance (e.g., BS 4449, NIS EN 197-1)
			Billing unit

Table: **locations** (Geopolitical & Economic Hubs)

Standardized geo-economic hubs capturing regional freight, distribution, and supply dynamics.

Column	Data Type	Modifiers / Constraints	Description
id	TEXT (UUID)	PRIMARY KEY	Unique location identifier
city	TEXT	NOT NULL	City or metropolis (e.g., Lagos, Abuja, Kano)
zone	TEXT	NOT NULL	Geopolitical Zone (e.g., SOUTH_WEST , NORTH_CENTRAL)
createdAt	TIMESTAMP(3)	Default: CURRENT_TIMESTAMP	Inception timestamp

Table: `master_materials` & `material_aliases` (Semantic Ingestion Engine)

AI-powered semantic backbone reconciling disparate trade names into unified standard codes.

Table	Column	Type	Details
<code>master_materials</code>	<code>id</code>	TEXT (UUID)	Primary Key
<code>master_materials</code>	<code>standardName</code>	TEXT	Unique standardized nomenclature
<code>master_materials</code>	<code>materialCode</code>	TEXT	Unique enterprise SKU / Uniclass / MasterFormat code
<code>master_materials</code>	<code>embedding</code>	FLOAT[]	768-dimensional vector from <code>text-embedding-004</code>
		TEXT	

Table: `unit_rate_items` & `unit_rate_components` (Parametric Cost Modeling)

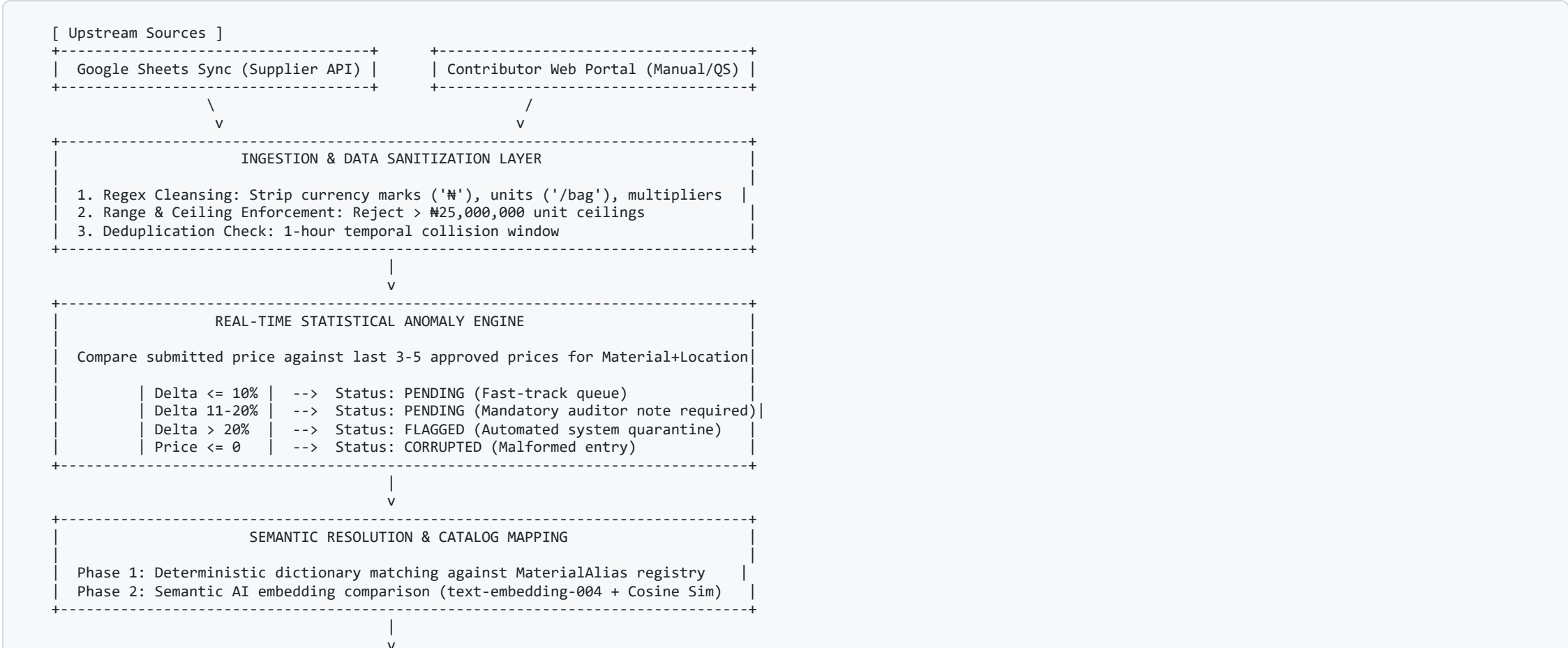
Enables algorithmic calculation of composite construction costs (e.g., *1m³ Reinforced Concrete Grade 25 in foundations*).

$$\text{Unit Rate} = (\text{Material Cost} + \text{Labor Cost} + \text{Plant Cost}) \times (1 + \text{Overhead \%}) \times (1 + \text{Profit \%})$$

Column	Table	Type	Purpose
<code>code</code>	<code>unit_rate_items</code>	<code>TEXT</code> (UK)	Standard BOQ item code (e.g., <code>CON-C25-01</code>)
<code>overheadPct</code>	<code>unit_rate_items</code>	<code>DECIMAL(5,2)</code>	Contractor overhead factor (default <code>15.00%</code>)
<code>profitPct</code>	<code>unit_rate_items</code>	<code>DECIMAL(5,2)</code>	Contractor profit margin (default <code>10.00%</code>)
			Component classification

3. Data Ingestion & Regional Rate Normalization Pipeline

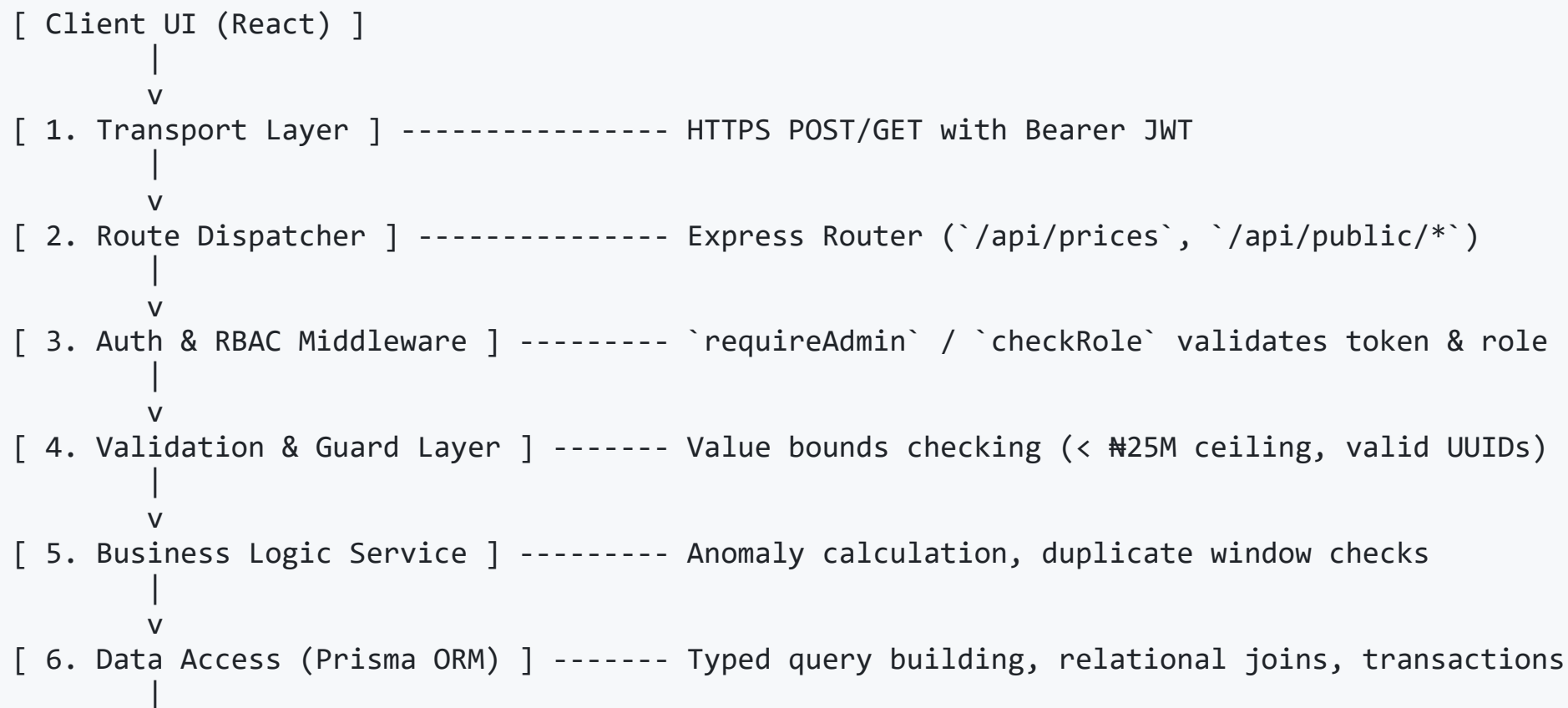
The ingestion pipeline handles heterogeneous data sources ranging from automated spreadsheets and field QS surveys to structured tender returns.



4. API & Data Flow Lifecycle

4.1 End-to-End Request Pipeline

A standard client request navigates through an eight-stage lifecycle:



4.2 Standard API Payloads

Sample 1: Ingestion / Price Submission Endpoint

- **Endpoint:** `POST /api/prices`
- **Access Clearance:** Authenticated (`SUPER_ADMIN` , `CONTRIBUTOR_QS`)
- **Headers:** `Authorization: Bearer <jwt>` , `Content-Type: application/json`

Request Payload:

```
{
  "materialId": "8f88921a-4d92-4217-91ba-c25e24340d85",
  "locationId": "e1f13b19-2169-425b-8032-fb9f82d24b61",
  "price": 10850.00,
  "currency": "NGN",
  "sourceType": "TENDER_RETURN",
  "notes": "Contractor returned bid for commercial high-rise foundation package, Victoria Island.",
  "contributorId": "+2348031112222 - Adekunle QS (Ibadan) - ade.qs@nccdb.com"
}
```

Response Payload (`201 Created`):

Sample 2: Public Regional Cost Intelligence Endpoint

- **Endpoint:** GET /api/public/analytics/regional/8f88921a-4d92-4217-91ba-c25e24340d85
- **Access Clearance:** Public (Open)

Request: No payload required (Target Material UUID in URL path).

Response Payload (200 OK):

```
[
  {
    "zone": "SOUTH_WEST",
    "averagePrice": 10625.50,
    "minPrice": 10500.00,
    "maxPrice": 10900.00,
    "dataPoints": 42
  },
  {
    "zone": "NORTH_CENTRAL",
    "averagePrice": 11550.00,
    "minPrice": 11200.00,
    "maxPrice": 11800.00,
    "dataPoints": 28
  },
  {
    "zone": "SOUTH_SOUTH",
    "averagePrice": 11340.00,
    "minPrice": 11000.00,
    "maxPrice": 11650.00,
    "dataPoints": 31
  }
]
```

5. Data Integrity, Scaling & Auditability

5.1 Referential Integrity Strategy

PostgreSQL foreign keys enforce strict lifecycle rules to prevent corrupted links:

- **Controlled Cascades:** Deleting a `Material` cascades to its `PriceRecord` and `MaterialSupplier` entries. Deleting a `UnitRateItem` cascades to its `UnitRateComponent` composition items.
- **Deletion Restrictions (`RESTRICT`):** Deleting a `Location` or `MaterialCategory` is blocked if any active `PriceRecord` or `Material` references it, preserving market integrity.
- **Audit Nullification (`SET NULL`):** If a user or auditor account is removed, historical `submittedById` and `approvedById` fields in `price_records` are set to `NULL`, retaining historical price logs.

Conclusion & Architectural Summary

The **NCCDB** platform combines strict relational integrity, statistical quality controls, and AI-driven normalization. By pairing PostgreSQL and Prisma with Google Workspace sync, Google GenAI embeddings, and responsive client visualizations, NCCDB delivers a scalable, auditable, and authoritative construction cost intelligence platform for Nigeria and emerging African markets.